

Research Methodology Report – Guidelines

The Research Methodology Report (Activity 3) should demonstrate your understanding of research methods relevant to your MRes project and justify the methodological choices you will make. In essence, the report should extend your Research Planning Report (Activity 4) primarily by adding **why particular methods are appropriate, how they will be applied, and how they compare with reasonable alternatives**.

Do not simply describe methods. Demonstrate critical methodological thinking by justifying choices, considering alternatives and limitations, and showing a clear connection between the research questions, methods, evidence and evaluation.

Suggested Structure

1. Introduction (~1.5 pages)

Briefly introduce the research problem, its significance, and the specific research question(s) or objectives. Clearly identify what the research seeks to discover, develop, demonstrate, compare, or evaluate. Keep general background concise and include only material needed to motivate the methodology.

2. Methodological Approach and Rationale (~2 pages)

Describe the overall research approach and the specific methods to be used. Explain why the methods are appropriate for answering the research questions. Discuss credible alternative approaches and explain why the selected approach is more suitable for your project.

3. Research Design and Implementation (~2–3 pages)

Explain how the research will actually be conducted. Depending on the project, this may include:

- experimental design, system/prototype development, algorithms or models;
- datasets, data collection and sampling;
- hardware/software platforms and research tools;
- experimental variables, controls, baselines and parameter settings; and
- training/testing procedures, simulations, surveys, or other techniques.

Present the work as a logical sequence of stages and explain how they fit together. A workflow or methodology diagram is encouraged where appropriate.

4. Evaluation and Data Analysis (~1–2 pages)

Explain how you will determine whether the proposed approach is successful. Identify appropriate metrics, baselines, experimental comparisons, statistical/analytical methods, validation procedures, and criteria for interpreting results. Where applicable, discuss reproducibility, reliability and validity. Evaluation should be directly linked to the research questions and expected contributions.

5. Limitations, Risks and Research Considerations (~0.5 page)

Critically discuss foreseeable limitations, assumptions, threats to validity, sources of bias or error, and practical risks, together with appropriate mitigation strategies. Where relevant, address ethics, safety, privacy, data management and responsible research considerations.

6. Summary (~0.5 page)

Summarise the methodology and explain how the selected methods collectively address the research objectives. Highlight expected outcomes, methodological strengths and remaining limitations.

References